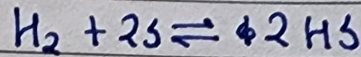
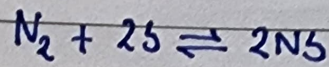


Q.1) Both N_2 and H_2 dissociate on adsorption, so each molecule needs a pair of adjacent vacant sites:



$$K_{N_2} = \frac{C_{NS}^2}{P_{N_2} C_v^2} \Rightarrow C_{NS} = C_v \sqrt{K_{N_2} P_{N_2}}$$

Similarly, $C_{HS} = C_v \sqrt{K_{H_2} P_{H_2}}$

Site balance

$$C_t = C_v + 2C_{NS} + 2C_{HS}$$

$$C_t = C_v \left(1 + 2\sqrt{K_{N_2} P_{N_2}} + 2\sqrt{K_{H_2} P_{H_2}} \right)$$

Replacing $C_{NS} / \sqrt{K_{N_2} P_{N_2}} = C_v$

$$\Rightarrow \frac{C_t \sqrt{K_{N_2} P_{N_2}}}{1 + 2\sqrt{K_{N_2} P_{N_2}} + 2\sqrt{K_{H_2} P_{H_2}}} = C_{NS}$$